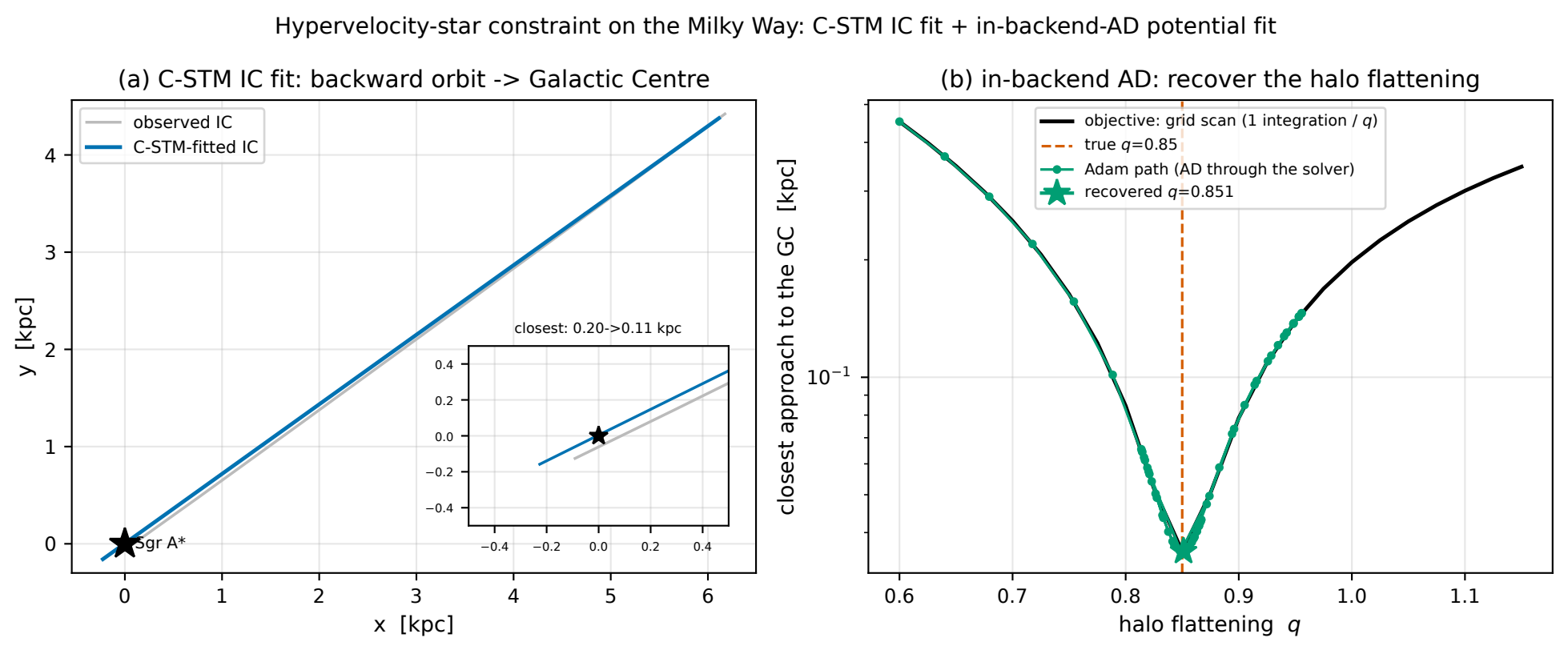
Hypervelocity-star constraint on the Milky Way: C-STM IC fit + in-backend-AD potential fit
(a) C-STM IC fit: backward orbit -> Galactic Centre
observed IC
C-STM-fitted IC
Sgr A*
closest: 0.20->0.11 kpc
y [kpc]
x [kpc]
(b) in-backend AD: recover the halo flattening
objective: grid scan (1 integration / q)
true q=0.85
Adam path (AD through the solver)
recovered q=0.851
closest approach to the GC [kpc]
halo flattening q